Induced Subgraphs in Ramsey Graphs

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Abstract

Replace this paragraph with a concise summary (100–200 words) of the dissertation: the problem studied, the methods used, and the principal results obtained.

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1 Introduction

2 Subgraph with Many Degrees

Definition 2.1 ((c, δ)-diversity)**.**

Theorem 2.2 (Bukh–Sudakov 2007 []). *Every C-Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(n)$ distinct degrees.*

Proof. □

3 The Weak Erdős–McKay Conjecture

Definition 3.1 ((δ, ε)-rich graph)**.**

Theorem 3.2 (Kwan–Sudakov 2020 []). *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Proof. □

4 The Erdős–McKay Conjecture

4.1 The Full Erdős–McKay Conjecture and Its Resolution

4.2 Quadratic Polynomials of Gaussians

4.3 The Carbery–Wright Theorem

Theorem 4.1 (Carbery–Wright []).

4.4 Fourier Analysis and Characteristic Functions

4.5 Additive Structure Dichotomy

4.6 The Two-Scale Decomposition

4.7 The Buckets Argument

Theorem 4.2 (Kwan–Sah–Sauermann–Sawhney 2023 []).

5 Conclusion and Future Work

References

A Supplementary Proofs